KENALILY SCHOOL FOR CHILDREN
An Udgamverse School

Computer Science Project

Project Based On

COFFEE ROASTERY INVENTORY & WHOLESALE BILLING SYSTEM WITH PYTHON & MYSQL

MySQL

Respected Teacher
MS. VRUNDA PATEL

Class
XII D

Student Name
RISHI PANDYA

Roll Number
24





The Idea Behind

The Problem

Specialty coffee roasteries source green coffee beans from different estates, roast them in small batches (Light, Medium, Dark), and supply them in bulk to cafes. Tracking remaining kilograms per batch and calculating bulk tier discounts manually in notebooks or spreadsheets often leads to stock mismatches and billing errors.

The Solution

The Coffee Roastery Inventory & Wholesale Billing System is a menu-driven application built using Python and MySQL. It acts as a central ledger that tracks live roasted coffee stock in kilograms, automates wholesale billing for client cafes, and updates inventory in real time.



To be Provided by the school

KENALILY SCHOOL FOR CHILDREN
An Udgamverse School

Computer Science Project

ACKNOWLEDGEMENT

CBSE Class XII Computer Science Practical (083)

Respected Teacher
MS. VRUNDA PATEL

Class
XII D
Session: 2026–27

Student Name
RISHI PANDYA

Roll No.
24

I would like to express my sincere gratitude to my Computer Science teacher, **Ms. Vrunda Patel**, for her vital guidance, encouragement, and continuous support throughout the development of this project, "**Coffee Roastery & Wholesale Billing System.**" Her clear explanations of Python–MySQL connectivity and relational database concepts made this practical implementation possible. I also extend my heartfelt thanks to our **respected Principal and the school administration** for providing the computer laboratory facilities and resources required to complete this work. Finally, I am grateful to **my parents** and for their constant support during the completion of this CBSE Class XII practical file.

Rishi Pandya
Class XII D | Roll No. 24



Features & Specs

Key Features

Live Batch Inventory: Tracks estate name, processing method (Washed, Natural, Honey), roast profile, and live stock in kilograms.

Automated Wholesale Discounts: Automatically applies a 10% discount on orders of 10 kg+ and a 15% discount on orders of 25 kg+ during billing.

Real-Time Stock Sync: Deducts ordered kilograms from the database immediately upon billing and restores stock automatically if an order is cancelled.

Roastery Analytics: Uses SQL aggregate functions (COUNT and SUM) to display total shipments, total kilograms sold, and net wholesale revenue.

System Requirements

Hardware: AMD Ryzen / Intel Core Processor, 4 GB+ RAM, 500 MB free storage.

Operating System: Windows 10 / Windows 11 (64-bit).

Frontend / Logic: Python 3.x (IDLE / VS Code).

Backend Database: MySQL Server 8.0 & mysql-connector-python.

MySQL



Database Design

Table 1: Batches

Stores roasted coffee lot details and live inventory in kilograms (roastery_db).

- batch_id (INT, Primary Key): Unique batch code (e.g., 101, 102).
- estate_name (VARCHAR(40)): Coffee estate and origin region.
- process (VARCHAR(20)): Bean processing (Washed, Natural, Honey).
- roast_level (VARCHAR(20)): Roast profile (Light, Medium, Dark).
- stock_kg (INT): Live roasted inventory available in kilograms.
- price_per_kg (INT): Base wholesale rate per kilogram (in Rs.).



MySQL Database Design

SQL Commands

Table 1: Creating & Populating Batches

```sql
CREATE DATABASE roastery_db;
USE roastery_db;

CREATE TABLE batches (
    batch_id INT PRIMARY KEY,
    estate_name VARCHAR(40),
    process VARCHAR(20),
    roast_level VARCHAR(20),
    stock_kg INT,
    price_per_kg INT
);

INSERT INTO batches VALUES (101, 'Riverdale Estate', 'Washed', 'Light-Medium', 80, 1400);
INSERT INTO batches VALUES (102, 'Ratnagiri Estate', 'Natural', 'Medium', 65, 1350);
INSERT INTO batches VALUES (103, 'Attikan Estate', 'Washed', 'Medium-Dark', 120, 1150);
INSERT INTO batches VALUES (104, 'Baarbara Estate', 'Honey', 'Medium', 50, 1250);
INSERT INTO batches VALUES (105, 'Malabar Estate', 'Monsooned', 'Dark', 90, 1050);
```

MySQL Output

Table 1: SQL Command Output

```
+----------+-------------------+-----------+--------------+----------+--------------+
| batch_id | estate_name       | process   | roast_level  | stock_kg | price_per_kg |
+----------+-------------------+-----------+--------------+----------+--------------+
|      101 | Riverdale Estate  | Washed    | Light-Medium |       80 |         1400 |
|      102 | Ratnagiri Estate  | Natural   | Medium       |       65 |         1350 |
|      103 | Attikan Estate    | Washed    | Medium-Dark  |      120 |         1150 |
|      104 | Baarbara Estate   | Honey     | Medium       |       50 |         1250 |
|      105 | Monsooned Malabar | Monsooned | Dark         |       90 |         1050 |
+----------+-------------------+-----------+--------------+----------+--------------+
5 rows in set (0.00 sec)
```

MySQL



Database Design

Table 2: Orders

Stores wholesale cafe invoices linked to batches via batch_id (roastery_db).

- order_id (INT, Primary Key): Unique wholesale invoice number.
- batch_id (INT, Foreign Key): Links the order to batches(batch_id).
- cafe_name (VARCHAR(40)): Name of the purchasing client cafe.
- kg_ordered (INT): Bulk quantity ordered in kilograms.
- discount_amt (FLOAT): Tier discount applied (10% on 10kg+, 15% on 25kg+).
- final_bill (FLOAT): Net payable wholesale invoice amount (in Rs.).



MySQL Database Design

SQL Commands

Table 2: Creating & Populating Orders

```sql
CREATE TABLE orders (
    order_id INT PRIMARY KEY,
    batch_id INT,
    cafe_name VARCHAR(40),
    kg_ordered INT,
    discount_amt FLOAT,
    final_bill FLOAT
);

INSERT INTO orders VALUES (501, 101, 'Third Wave Cafe', 15, 2100, 18900);
INSERT INTO orders VALUES (502, 103, 'Blue Tokai Roasters', 30, 5175, 29325);
INSERT INTO orders VALUES (503, 102, 'Dyu Art Cafe', 8, 0, 10800);
INSERT INTO orders VALUES (504, 104, 'Project Hum', 12, 1800, 13200);
```

MySQL Output

Table 2: SQL Command Output

```
MySQL 8.4 LTS Command Line Client

+----------+----------+--------------------+------------+--------------+------------+
| order_id | batch_id | cafe_name          | kg_ordered | discount_amt | final_bill |
+----------+----------+--------------------+------------+--------------+------------+
|      501 |      101 | Third Wave Cafe    |         15 |         2100 |      18900 |
|      502 |      103 | Blue Tokai Roasters |         30 |         5175 |      29325 |
|      503 |      102 | Dyu Art Cafe       |          8 |            0 |      10800 |
|      504 |      104 | Project Hum        |         12 |         1800 |      13200 |
+----------+----------+--------------------+------------+--------------+------------+
4 rows in set (0.00 sec)
```



Python Source Code

Source Code: Lines (1-47)

```python
import mysql.connector

con = mysql.connector.connect(
    host="localhost", user="root", passwd="your_password", database="roastery_db"
)
cur = con.cursor()

while True:
  print("\n=========================================")
  print("   COFFEE ROASTERY & WHOLESALE SYSTEM")
  print("=========================================")
  print("1. View All Coffee Batches")
  print("2. Place New Wholesale Order")
  print("3. View All Wholesale Orders")
  print("4. Restock Coffee Batch (Add kg)")
  print("5. Cancel an Order")
  print("6. Exit")

  ch = int(input("\nEnter your choice (1-6): "))

  if ch == 1:
    print("\n--- AVAILABLE COFFEE BATCHES ---")
    cur.execute("SELECT * FROM batches")
    data = cur.fetchall()
    for row in data:
      print(
          "Batch ID:",
          row[0],
          "| Estate:",
          row[1],
          "| Process:",
          row[2],
          "| Roast:",
          row[3],
          "| Stock:",
          row[4],
          "kg | Price: Rs.",
          row[5],
      )

  elif ch == 2:
    o_id = int(input("Enter New Order ID (e.g. 505): "))
    b_id = int(input("Enter Batch ID to buy (101-105): "))
    cafe = input("Enter Buyer Cafe Name: ")
    kg = int(input("Enter Quantity in kg: "))
    rate = int(input("Enter Price per kg of that Batch: "))

```

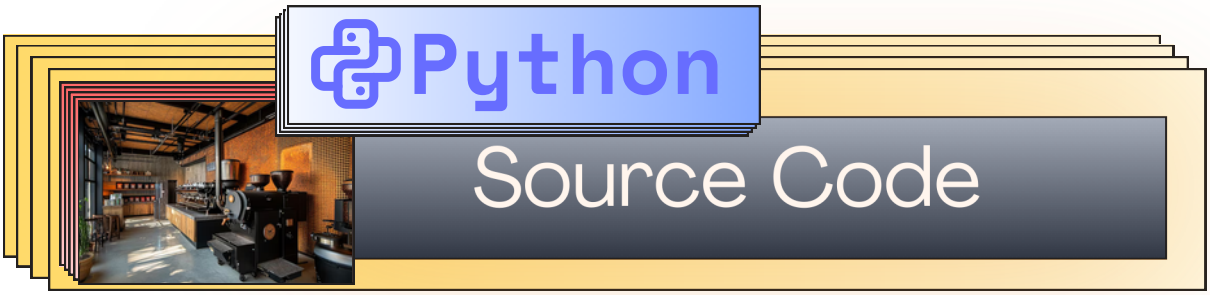


Python

Source Code

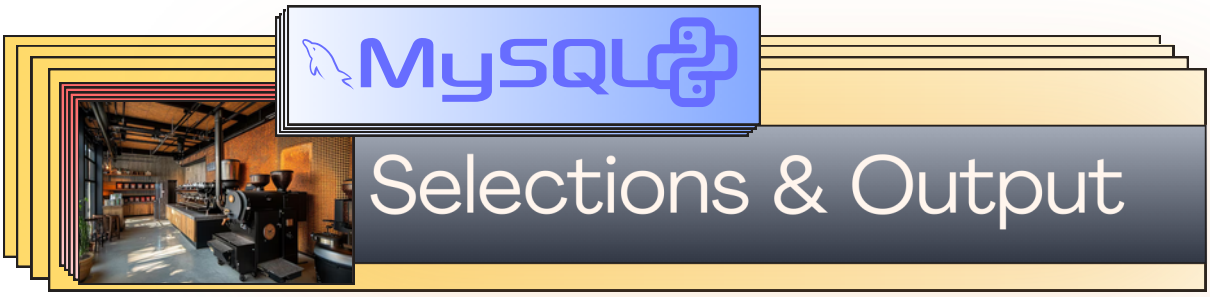
Source Code: Lines (48-94)

```
48      gross = kg * rate
49      if kg >= 25:
50        disc = gross * 0.15
51      elif kg >= 10:
52        disc = gross * 0.10
53      else:
54        disc = 0
55
56      final_bill = gross - disc
57
58      q1 = "INSERT INTO orders VALUES ({}, {}, '{}', {}, {}, {})".format(
59          o_id, b_id, cafe, kg, disc, final_bill
60      )
61      cur.execute(q1)
62
63      q2 = "UPDATE batches SET stock_kg = stock_kg - {} WHERE batch_id = {}".format(
64          kg, b_id
65      )
66      cur.execute(q2)
67      con.commit()
68
69      print("\n--- WHOLESALE BILL GENERATED ---")
70      print("Cafe Name     :", cafe)
71      print("Gross Amount  : Rs.", gross)
72      print("Bulk Discount : Rs.", disc)
73      print("Final Bill    : Rs.", final_bill)
74
75  elif ch == 3:
76      print("\n--- ALL WHOLESALE ORDERS ---")
77      cur.execute("SELECT * FROM orders")
78      data = cur.fetchall()
79      for row in data:
80        print(
81            "Order ID:",
82            row[0],
83            "| Batch ID:",
84            row[1],
85            "| Cafe:",
86            row[2],
87            "| Qty:",
88            row[3],
89            "kg | Discount: Rs.",
90            row[4],
91            "| Total: Rs.",
92            row[5],
93        )
94
```



Source Code: Lines (95-119)

```
 95  elif ch == 4:
 96    b_id = int(input("Enter Batch ID to restock: "))
 97    add_kg = int(input("Enter newly roasted kg to add: "))
 98
 99    q = "UPDATE batches SET stock_kg = stock_kg + {} WHERE batch_id = {}".format(
100        add_kg, b_id
101    )
102    cur.execute(q)
103    con.commit()
104    print("Stock updated successfully!")
105
106  elif ch == 5:
107    o_id = int(input("Enter Order ID to cancel: "))
108
109    q = "DELETE FROM orders WHERE order_id = {}".format(o_id)
110    cur.execute(q)
111    con.commit()
112    print("Order cancelled and deleted from records!")
113
114  elif ch == 6:
115    print("Thank you! Closing system.")
116    break
117
118  else:
119    print("Invalid choice! Please enter 1 to 6.")
```



Output 1: Main Menu Interface

```
roastery.py - Python 3.12 Shell

==================== RESTART: C:/Users/User/Desktop/roastery.py ====================

==========================================
   COFFEE ROASTERY & WHOLESALE SYSTEM
==========================================
1. View All Coffee Batches
2. Place New Wholesale Order
3. View All Wholesale Orders
4. Restock Coffee Batch (Add kg)
5. Cancel an Order
6. Exit

Enter your choice (1-6):
```

Choice 1: View All Coffee Batches

```
roastery.py - Python 3.12 Shell

==================== RESTART: C:/Users/User/Desktop/roastery.py ====================

==========================================
   COFFEE ROASTERY & WHOLESALE SYSTEM
==========================================
1. View All Coffee Batches
2. Place New Wholesale Order
3. View All Wholesale Orders
4. Restock Coffee Batch (Add kg)
5. Cancel an Order
6. Exit

Enter your choice (1-6): 1

--- AVAILABLE COFFEE BATCHES ---
Batch ID: 101 | Estate: Riverdale Estate | Process: Washed | Roast: Light-Medium | Stock: 80 kg | Price: Rs. 1400
Batch ID: 102 | Estate: Ratnagiri Estate | Process: Natural | Roast: Medium | Stock: 65 kg | Price: Rs. 1350
Batch ID: 103 | Estate: Attikan Estate | Process: Washed | Roast: Medium-Dark | Stock: 120 kg | Price: Rs. 1150
Batch ID: 104 | Estate: Baarbara Estate | Process: Honey | Roast: Medium | Stock: 50 kg | Price: Rs. 1250
Batch ID: 105 | Estate: Monsooned Malabar | Process: Monsooned | Roast: Dark | Stock: 90 kg | Price: Rs. 1050
```



MySQL Selections & Output

Choice 2: Place New Wholesale Order

```
roastery.py - Python 3.12 Shell

================== RESTART: C:/Users/User/Desktop/roastery.py ==================

==========================================
   COFFEE ROASTERY & WHOLESALE SYSTEM
==========================================
1. View All Coffee Batches
2. Place New Wholesale Order
3. View All Wholesale Orders
4. Restock Coffee Batch (Add kg)
5. Cancel an Order
6. Exit

Enter your choice (1-6): 2
Enter New Order ID (e.g. 505): 505
Enter Batch ID to buy (101-105): 101
Enter Buyer Cafe Name: Roastery Coffee House
Enter Quantity in kg: 20
Enter Price per kg of that Batch: 1400

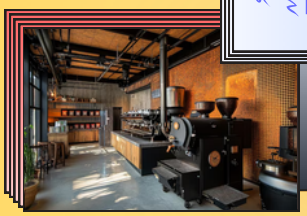
--- WHOLESALE BILL GENERATED ---
Cafe Name      : Roastery Coffee House
Gross Amount   : Rs. 28000
Bulk Discount : Rs. 2800.0
Final Bill     : Rs. 25200.0
```

MySQL Table After Choice 2

(Command used: SELECT * FROM orders;)

```
MySQL 8.4 LTS Command Line Client

+----------+----------+----------------------+------------+--------------+------------+
| order_id | batch_id | cafe_name            | kg_ordered | discount_amt | final_bill |
+----------+----------+----------------------+------------+--------------+------------+
|      501 |      101 | Third Wave Cafe      |         15 |         2100 |      18900 |
|      502 |      103 | Blue Tokai Roasters  |         30 |         5175 |      29325 |
|      503 |      102 | Dyu Art Cafe         |          8 |            0 |      10800 |
|      504 |      104 | Project Hum          |         12 |         1800 |      13200 |
|      505 |      101 | Roastery Coffee House |         20 |         2800 |      25200 |
+----------+----------+----------------------+------------+--------------+------------+
5 rows in set (0.00 sec)
```



MySQL Selections & Output

Choice 3: View All Wholesale Orders

```
roastery.py - Python 3.12 Shell

========== RESTART: C:/Users/User/Desktop/roastery.py ==========

==========================================
   COFFEE ROASTERY & WHOLESALE SYSTEM
==========================================
1. View All Coffee Batches
2. Place New Wholesale Order
3. View All Wholesale Orders
4. Restock Coffee Batch (Add kg)
5. Cancel an Order
6. Exit

Enter your choice (1-6): 3

--- ALL WHOLESALE ORDERS ---
Order ID: 501 | Batch ID: 101 | Cafe: Third Wave Cafe | Qty: 15 kg | Discount: Rs. 2100.0 | Total: Rs. 18900.0
Order ID: 502 | Batch ID: 103 | Cafe: Blue Tokai Roasters | Qty: 30 kg | Discount: Rs. 5175.0 | Total: Rs. 29325.0
Order ID: 503 | Batch ID: 102 | Cafe: Dyu Art Cafe | Qty: 8 kg | Discount: Rs. 0.0 | Total: Rs. 10800.0
Order ID: 504 | Batch ID: 104 | Cafe: Project Hum | Qty: 12 kg | Discount: Rs. 1800.0 | Total: Rs. 13200.0
Order ID: 505 | Batch ID: 101 | Cafe: Roastery Coffee House | Qty: 20 kg | Discount: Rs. 2800.0 | Total: Rs. 25200.0
```

MySQL Table Verification for Choice 3

(Command used: SELECT * FROM orders;)

```
MySQL 8.4 LTS Command Line Client

+----------+----------+----------------------+------------+--------------+------------+
| order_id | batch_id | cafe_name            | kg_ordered | discount_amt | final_bill |
+----------+----------+----------------------+------------+--------------+------------+
|      501 |      101 | Third Wave Cafe      |         15 |         2100 |      18900 |
|      502 |      103 | Blue Tokai Roasters  |         30 |         5175 |      29325 |
|      503 |      102 | Dyu Art Cafe         |          8 |            0 |      10800 |
|      504 |      104 | Project Hum          |         12 |         1800 |      13200 |
|      505 |      101 | Roastery Coffee House |         20 |         2800 |      25200 |
+----------+----------+----------------------+------------+--------------+------------+
5 rows in set (0.00 sec)
```



Selections & Output

Choice 4: Restock Coffee Batch (Kg)

roastery.py - Python 3.12 Shell

```
================== RESTART: C:/Users/User/Desktop/roastery.py ==================

==========================================
   COFFEE ROASTERY & WHOLESALE SYSTEM
==========================================
1. View All Coffee Batches
2. Place New Wholesale Order
3. View All Wholesale Orders
4. Restock Coffee Batch (Add kg)
5. Cancel an Order
6. Exit

Enter your choice (1-6): 4
Enter Batch ID to restock: 101
Enter newly roasted kg to add: 40
Stock updated successfully!
```

MySQL Table After Choice 4

(Command used: SELECT * FROM batches;)

MySQL 8.4 LTS Command Line Client

```
+----------+--------------------+-----------+--------------+----------+--------------+
| batch_id | estate_name        | process   | roast_level  | stock_kg | price_per_kg |
+----------+--------------------+-----------+--------------+----------+--------------+
|      101 | Riverdale Estate   | Washed    | Light-Medium |      100 |         1400 |
|      102 | Ratnagiri Estate   | Natural   | Medium       |       65 |         1350 |
|      103 | Attikan Estate     | Washed    | Medium-Dark  |      120 |         1150 |
|      104 | Baarbara Estate    | Honey     | Medium       |       50 |         1250 |
|      105 | Monsooned Malabar  | Monsooned | Dark         |       90 |         1050 |
+----------+--------------------+-----------+--------------+----------+--------------+
5 rows in set (0.00 sec)
```



MySQL Selections & Output

Choices 5 & 6: Cancel Order & Exit

roastery.py - Python 3.12 Shell

```
==================== RESTART: C:/Users/User/Desktop/roastery.py ====================

==========================================
   COFFEE ROASTERY & WHOLESALE SYSTEM
==========================================
1. View All Coffee Batches
2. Place New Wholesale Order
3. View All Wholesale Orders
4. Restock Coffee Batch (Add kg)
5. Cancel an Order
6. Exit

Enter your choice (1-6): 5
Enter Order ID to cancel: 503
Order cancelled and deleted from records!

==========================================
   COFFEE ROASTERY & WHOLESALE SYSTEM
==========================================
1. View All Coffee Batches
2. Place New Wholesale Order
3. View All Wholesale Orders
4. Restock Coffee Batch (Add kg)
5. Cancel an Order
6. Exit

Enter your choice (1-6): 6
Thank you! Closing system.
```

MySQL Table After Choice 5

(Command used: SELECT * FROM orders;)

MySQL 8.4 LTS Command Line Client

order_id	batch_id	cafe_name	kg_ordered	discount_amt	final_bill
501	101	Third Wave Cafe	15	2100	18900
502	103	Blue Tokai Roasters	30	5175	29325
504	104	Project Hum	12	1800	13200
505	101	Roastery Coffee House	20	2800	25200

4 rows in set (0.00 sec)



System Limitations

Functional Limitations

Command-Line Interface (CLI): The software operates strictly within a text terminal, lacking a graphical window (GUI) for mouse clicks, drop-down menus, and visual buttons.

Manual Identity Input: Order IDs and Batch IDs must be entered manually by the user, requiring prior verification to avoid entering duplicate primary keys.

Single-Session Workflow: The program is designed for single-operator use at the roastery counter and does not support simultaneous multi-terminal cashier sessions.

Basic Billing Formatting: Generates text-based console invoices rather than printable PDF or thermal POS receipts.

Technical Limitations

Unencrypted In-Script Credentials: Database authentication details (host, root username, and password) are hardcoded directly within the Python script in plain text.

Strict Type Handling: The script relies on direct type casting (int(), float()); entering non-numeric characters for batch IDs or kilogram amounts halts execution due to unhandled ValueError exceptions



References & Bibliography

Academic References & Textbooks

Computer Science with Python (Class XII): Preeti Arora, Sultan Chand Educational Publishers, New Delhi.

Computer Science with Python (Class XII): Sumita Arora, Dhanpat Rai & Co., New Delhi.

NCERT Computer Science Textbook (Class XII): National Council of Educational Research and Training, New Delhi.

CBSE Curriculum & Practical Guidelines (Subject Code 083): Central Board of Secondary Education, New Delhi.

Digital Documentation & Web Resources

Official Python Documentation (v3.12): https://docs.python.org/3/ (Standard functions, control flow, and exception handling).

MySQL 8.4 Reference Manual: https://dev.mysql.com/doc/ (Relational schema design, constraint handling, and SQL syntax).

MySQL Connector/Python Developer Guide: Official Oracle driver reference for database cursor interactions and connection management.

W3Schools & GeeksforGeeks Python-SQL Tutorials: Reference guides for tabular console formatting and SQL query patterns.



Digital Files

Through Link or QR Code

Assets Direct Download (.zip):
https://codeload.github.com/RishixD4511/The-Roastery/zip/refs/heads/main

Assets Folder (Google Drive Folder):
https://drive.google.com/drive/folders/1OtJ57bBOzFttPxcfi8zxApqVN8ZcNZ0V?usp=sharing

Physical USB Drive



The End.
-fin-